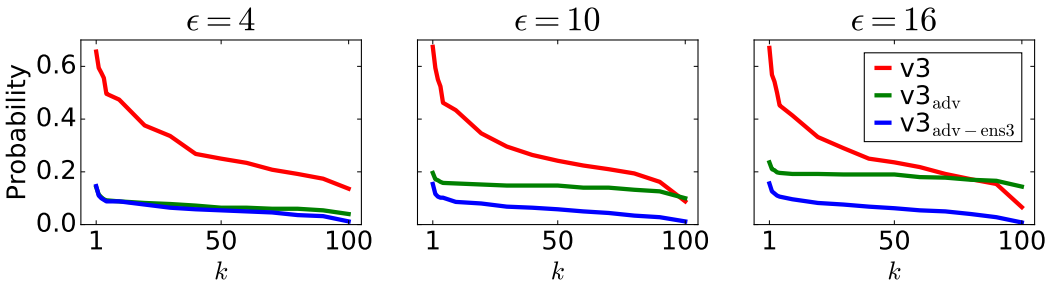

Figure 2: **The dimensionality of the adversarial cone**. For 500 correctly classified points x, and for $\epsilon \in \{4, 10, 16\}$, we plot the probability that we find *at least* k orthogonal vectors r_i such that $\|r_i\|_\infty = \epsilon$ and $x + r_i$ is misclassified. For $\epsilon \geq 10$, model v3_{adv} shows a *bimodal* phenomenon: most points x either have 0 adversarial directions or more than 90.

(we recall that for models v3 and v3_{adv}, the approximation ratio was respectively 19% and 7%). Similarly, for the $\text{IRv2}_{\text{adv-ens}}$ model, the ratio improves from 8% (for IRv2_{adv}) to 14%. As expected, not solely training on a white-box single-step attack reduces gradient masking. We also verify that after Ensemble Adversarial Training, a two-step iterative attack outperforms the R+Step-LL attack from Section 4.1, thus providing further evidence that these models have meaningful gradients.

Finally, we revisit the "Gradient-Aligned Adversarial Subspace" (GAAS) method of Tramèr et al. (2017). Their method estimates the size of the space of adversarial examples in the vicinity of a point, by finding a set of *orthogonal* perturbations of norm ϵ that are all adversarial. We note that adversarial perturbations do not technically form a "subspace" (e.g., the $\mathbf{0}$ vector is not adversarial). Rather, they may form a "cone", the dimension of which varies as we increase ϵ. By linearizing the loss function, estimating the dimensionality of this cone reduces to finding vectors r_i that are strongly aligned with the model's gradient $g = \nabla_x L(h(x), y_{\text{true}})$. Tramèr et al. (2017) give a method that finds k orthogonal vectors r_i that satisfy $g^\top r_i \geq \epsilon \cdot \|g\|_2 \cdot \frac{1}{\sqrt{k}}$ (this bound is tight). We extend this result to the ℓ_∞ norm, an open question in Tramèr et al. (2017). In Section E, we give a randomized combinatorial construction (Colbourn, 2010), that finds k orthogonal vectors r_i satisfying $\|r_i\|_\infty = \epsilon$ and $\mathbb{E}\left[g^\top r_i\right] \geq \epsilon \cdot \|g\|_1 \cdot \frac{1}{\sqrt{k}}$. We show that this result is tight as well.

For models v3, v3_{adv} and $\text{v3}_{\text{adv-ens3}}$, we select 500 correctly classified test points. For each x, we search for a maximal number of orthogonal adversarial perturbations r_i with $\|r_i\|_\infty = \epsilon$. We limit our search to $k \leq 100$ directions per point. The results are in Figure 2. For $\epsilon \in \{4, 10, 16\}$, we plot the proportion of points that have at least k orthogonal adversarial perturbations. For a fixed ϵ, the value of k can be interpreted as the dimension of a "slice" of the cone of adversarial examples near a data point. For the standard Inception v3 model, we find over 50 orthogonal adversarial directions for 30% of the points. The v3_{adv} model shows a curious bimodal phenomenon for $\epsilon \geq 10$: for most points ($\approx 80\%$), we find no adversarial direction aligned with the gradient, which is consistent with the gradient masking effect. Yet, for most of the remaining points, the adversarial space is very high-dimensional ($k \geq 90$). Ensemble Adversarial Training yields a more robust model, with only a small fraction of points near a large adversarial space.

5 Conclusion and Future Work

Previous work on adversarial training at scale has produced encouraging results, showing strong robustness to (single-step) adversarial examples (Goodfellow et al., 2014b; Kurakin et al., 2017b). Yet, these results are misleading, as the adversarially trained models remain vulnerable to simple black-box and white-box attacks. Our results, generic with respect to the application domain, suggest that adversarial training can be improved by *decoupling* the generation of adversarial examples from the model being trained. Our experiments with Ensemble Adversarial Training show that the robustness attained to attacks from some models *transfers* to attacks from other models.

We did not consider black-box adversaries that attack a model via other means than by transferring examples from a local model. For instance, generative techniques (Baluja & Fischer, 2017) might provide an avenue for stronger attacks. Yet, a recent work by Xiao et al. (2018) found Ensemble

Adversarial Training to be resilient to such attacks on MNIST and CIFAR10, and often attaining higher robustness than models that were adversarially trained on iterative attacks.

Moreover, *interactive* adversaries (see Appendix A) could try to exploit queries to the target model's prediction function in their attack, as demonstrated in Papernot et al. (2017). If queries to the target model yield *prediction confidences*, an adversary can estimate the target's gradient at a given point (e.g., using finite-differences as in Chen et al. (2017)) and fool the target with our R+FGSM attack. Note that if queries only return the predicted label, the attack does not apply. Exploring the impact of these classes of black-box attacks and evaluating their scalability to complex tasks is an interesting avenue for future work.

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